

Patrick M. TIMONS

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EDUCATION

Artificial Intelligence and Decision Making — *Bachelor of Science*
Massachusetts Institute of Technology

AUG 2021 - MAY 2025
GPA: 4.7/5.0

- *Software Engineering*: Software Performance Engineering, Algorithms II, Algorithms I, Computer Programming, Computation Structures, Low-Level Programming, Mathematics for Comp Sci
- *Mathematics and Machine Learning*: Inference and Information (G), Machine Learning (G), Machine Learning (UG), Deep Learning (G), Deep Learning (UG), Probability, Statistics, Linear Algebra and Optimization, Representation, Inference, and Reasoning in AI, Quantitative NLP, Computer Vision
- *Economics and Social Science*: Networks, Microeconomics, Macroeconomics, Information Policy
- *Other Science*: Linguistics, Chemistry, Biology

WORK EXPERIENCE

AWS Billing — *SDE*

SEP 2025 - PRESENT

- Utilized caching in an internal batch compute framework to launch Attribute Based Discounts in the ZHY partition
- Created infrastructure to support regression testing for strategic AWS customers undergoing major billing system changes
- Used Java, Batch Compute Framework, Protobuf, Apache Parquet, DuckDB, Build Tools, Guice, Spock, AWS CDK

Stealth Start-up — *Co-Founder, CTO*

SEP 2024 - MAR 2025

- MIT \$100K Finalist: Recognized among top MIT startups for innovation and market potential
- Winner of Harvard Law Entrepreneurship Competition: Gained access to mentorship from 8 Harvard Law School attorneys
- MIT Sandbox Grant Recipient: Awarded \$10,000 (\$5,000 cash and \$5,000 AWS Credits) for product development and scaling

AWS Billing — *SDE Intern*

JUN 2024 - AUG 2024

- Designed and implemented an end-to-end data pipeline to process real-time billing data of 550,000 enterprise AWS customers
- Built an ML-powered forecasting service for AWS cost-savings data analysis, served through a query-based API and a front-end dashboard
- Used AWS, Athena, IaC, CDK, SageMaker, S3, API Gateway, Data Catalog, Java, TypeScript, Python

TECHNICAL PROJECTS

Decaf Compiler

GitHub

- Building a compiler for Decaf programming language from scratch with Java (WIP)

Leiserchess

GitHub

- Optimized alpha-beta pruning implementation to create competitive Leiserchess engine

Improving Deep Learning Based Molecular Fingerprints Through Informed Resampling

GitHub

- Worked in a small team to research improvements to pretraining methods for transformer-based molecular encoders
- Created custom RoBERTa model through the use of transformers and bert-loves-chemistry (ChemBERTa) software packages with 9.1 percent improvement in Spearman’s rank correlation coefficient on downstream tasks compared to the base model
- Coauthored 5-page ACL-submission-style write-up about project and results

Recovering Latent Variables with Variational Autoencoders despite Training Bias

GitHub

- Researched how beta-regularization robustifies VAEs to training bias when attempting to recover latent variables
- Trained models with PyTorch Lightning and used scientific computing libraries to generate training data and visualize results

EXTRACURRICULAR ACTIVITIES AND HONORS

MIT Men’s Lacrosse — *Team Member*

AUG 2021 - MAY 2025

- Awarded USILA Academic All-American for the 2025 Season
- Received Ben Martin Award for “senior lacrosse player who has continuously exhibited the qualities of dedication, determination, and spirit”
- Voted Most Improved for 2023 season, NEWMAC All-Academic Award (2023, 2024, 2025), 2022 NEWMAC Champion

Global Teaching Labs — *Applied Math Teacher*

JAN 2024 - JAN 2024

- Taught selected topics in Operations Research with a focus on Bayesian Inference to high school students in Cremona, Italy

SKILLS

Foundational Data Structures & Algorithms, Programming (Java, Python, Bash, C, TypeScript)

Design Patterns and Programming Paradigms Model-first development, Dependency Injection, Object-Oriented-Programming, Procedural Programming, Concurrency, Multithreading

Dev Ops Source Control, Build Tools, Dependency Management, CI/CD

Databases SQL, AWS Data Catalog, Apache Parquet, DuckDB

Libraries and Ecosystems React, Amplify, AWS, SageMaker, PyTorch Lightning, NumPy, Pandas, scikit-learn, Transformers, Transformers Reinforcement Library, Matplotlib, Plotly, Dash